Jerry (Yuze) Li

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Education

University of Waterloo Expected 2029
Bachelor of Computer Science | Schulich Leader Scholarship ($100,000+) *Waterloo, Canada*

Experience

Mathos AI (YC W24) *Software Engineer Intern* June 2026 – Present

- Shipped MP4 export for AI math videos, rendering Remotion jobs in isolated AWS Fargate sandboxes via Celery.
- Built an OAuth 2.0 and MCP server so ChatGPT and Claude users can generate and download Mathos videos in chat.
- Shipped lifecycle emails on AWS SES and EventBridge with engagement tracking and Gmail reply threading.
- Rebuilt the AI tutor's step grading on a fast judge model plus SymPy checks, sharply improving accuracy and speed.

Waterloo Defence *Software Lead* Sept. 2026 – Present

- Lead software for a counter-drone perception system; built a detector (C motion front end, 3 MB CNN, tracker) that picks out distant drones in 4K video at 0.81 F1 on LRDDv3, 4× the classical baseline.
- Trained it on AWS GPUs over 370 GB of 4K video with fault-tolerant spot jobs; the 3 MB model scores within 2 AP50 of the best published RGB result on the FRED benchmark.

TopsOJ *Founder & Lead Developer* July 2023 – Present

- Built a competitive-math judge with 100,000+ all-time active users (Google Analytics), 180+ countries, and 128,000+ submissions.
- Designed EloMMR ratings; led a 30-person team across engineering, contest ops, and sponsors like 3Blue1Brown.

Quoted Tech Computers *Software Engineer Intern* July 2025 – Aug. 2025

- Automated barcode-label generation and CUPS printing with Node.js/Express, cutting manual processing time by 70%.
- Built Playwright/Selenium scrapers that structure competitor catalogs for pricing and inventory analysis.

Research

Recursive Delegation in Coding Agents *Independent Research, Sole Author* 2026

- **Accepted at two NeurIPS 2026 workshops:** Meta-Agents and Who Verifies the Agents? (unpublished).
- Ran controlled experiments on recursive coding-agent delegation, measuring verified task success, token cost, and overlapping code edits across 1–15-agent configurations.

Computational Astrophysics Nov. 2024 – Mar. 2026
Supervised by Dr. Alexander Mushtukov (University of Oxford) *Remote*

- Sole-authored papers in *Research Notes of the AAS* and *Journal of High School Science* (**Best Manuscript**), plus an arXiv preprint on binary-star mass transfer.
- Developed a velocity-rescaling closure that reduced orbit error from 22% to under 10^{-4}%.

Projects

Daedalus | *TypeScript, Three.js/WebGL, MediaPipe* JAMHacks 10

- Co-built a webcam-gesture 3D modeling app with MediaPipe hand tracking and Three.js; won Best Non-GenAI Hack.

Worldline Engine | *C++, raylib, CMake* 2026

- Wrote a deterministic C++ universe simulator with seeded generation, live physics, save/replay, and automated tests.

Awards

USACO Platinum (1,000/1,000 on Gold) | **AIME** qualifier (3×)
AAPT PhysicsBowl Div. II: 1st in Canada, 7th globally | **Sir Isaac Newton Exam**: 4th in Canada, 9th globally

Technical Skills

Languages: C/C++, Python, TypeScript, JavaScript, SQL
Frameworks: Node.js, Express, React, Next.js, Celery, PyTorch, ONNX Runtime, NumPy, SciPy, raylib, Three.js
Tools: AWS (EC2, Fargate, SES, S3), Git, Linux, SQLite, CMake, OpenCV, MediaPipe, Playwright, WebSockets